Multiplying by a One-Digit Number
Answer Key

Grade 4 — Number and Operations in Base Ten

Quick Answers

1. 252
2. 406
3. 3 x 200 = 600, 3 x 10 = 30, 3 x 4 = 12, total product = 642
4. 4 x 1000 = 4000, 4 x 300 = 1200, 4 x 20 = 80, 4 x 5 = 20, total product = 5300
5. 15335
6. (a) 412 rounded to the nearest hundred = 400, (b) the estimated product = 2400, (c) the exact product = 2472
7. 19320
8. (a) the exact number of seats = 8736 seats, (b) that total rounded to the nearest thousand = 9000
9. 51336
10. 6 x 3000 = 18000, 6 x 200 = 1200, 6 x 0 tens = 0, 6 x 8 = 48, total product = 19248
11. (a) 1875 rounded to the nearest thousand = 2000, (b) the estimated number of bricks = 8000, (c) the exact number of bricks = 7500 bricks, —
12. (a) the product = 8040, —

What is verified

10 of 12 problems fully machine-checked against SymPy.
— marks an answer only you can judge — problems 11, 12. The worked solution states what a correct response must contain.

Curriculum

Level: Grade 4 — Number and Operations in Base Ten
4.NBT.B.5 – *problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12*
Difficulty 1–5 of 5 across 30 tagged checks.

Common wrong answers

1. *If they got 242: wrote the ones digit but never carried the regrouped ten into the tens column*
5. *If they got 1835: dropped the zero out of the hundreds place and multiplied 367 by 5 instead*
9. *If they got 5166: treated the zero in the tens place as if it were not there and multiplied 574 by 9*

1. Multiply 63×4.

Solution: Ones: $4 \times 3 = 12$, so write 2 and carry 1 ten. Tens: $4 \times 6 = 24$ tens, plus the carried ten is 25 tens. $\boxed{252}$

2. Multiply 58×7.

Solution: Ones: $7 \times 8 = 56$, write 6 and carry 5 tens. Tens: $7 \times 5 = 35$ tens, plus 5 carried tens is 40 tens, which is 400. $\boxed{406}$

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3. Partial products for 214×3.

Solution: Split 214 by place value: $200 + 10 + 4$. Multiply each part by 3:

$$3 \times 200 = \boxed{600}, \qquad 3 \times 10 = \boxed{30}, \qquad 3 \times 4 = \boxed{12}$$

Adding the partial products: $600 + 30 + 12 = 642$. This is the same arithmetic the standard algorithm does, with the place values written out instead of carried. $\boxed{642}$

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4. Area model for $1{,}325 \times 4$.

Solution: $1{,}325 = 1000 + 300 + 20 + 5$, so the model has four parts:

$$4 \times 1000 = \boxed{4000}, \quad 4 \times 300 = \boxed{1200}, \quad 4 \times 20 = \boxed{80}, \quad 4 \times 5 = \boxed{20}$$

Total: $4000 + 1200 + 80 + 20 = 5300$. Adding the biggest part first keeps the place values lined up. $\boxed{5300}$

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5. Multiply $3{,}067 \times 5$.

Solution: Ones: $5 \times 7 = 35$; write 5, carry 3. Tens: $5 \times 6 = 30$, plus 3 is 33; write 3, carry 3. Hundreds: $5 \times 0 = 0$, plus the carried 3 is 3 — the zero digit still occupies its column, and the carry lands there. Thousands: $5 \times 3 = 15$. $\boxed{15335}$

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6. 412×6: (a) round, (b) estimate, (c) exact.

Solution (a): 412 is between 400 and 500, and its tens digit 1 is below 5, so it rounds down. $\boxed{400}$

Solution (b): $6 \times 400 = 2400$. $\boxed{2400}$

Solution (c): 6×412: ones $6 \times 2 = 12$, write 2 carry 1; tens $6 \times 1 = 6$, plus 1 is 7; hundreds $6 \times 4 = 24$. The exact answer is a little above the estimate, which is what you expect after rounding down. $\boxed{2472}$

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7. Multiply $2{,}415 \times 8$.

Solution: Ones: $8 \times 5 = 40$; write 0, carry 4. Tens: $8 \times 1 = 8$, plus 4 is 12; write 2, carry 1. Hundreds: $8 \times 4 = 32$, plus 1 is 33; write 3, carry 3. Thousands: $8 \times 2 = 16$, plus 3 is 19. $\boxed{19320}$

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8. Theatre: 1,248 seats, 7 sold-out shows. (a) Exact total. (b) Rounded to the nearest thousand.

Solution (a): 7×1248: ones $7 \times 8 = 56$, write 6 carry 5; tens $7 \times 4 = 28$, plus 5 is 33, write 3 carry 3; hundreds $7 \times 2 = 14$, plus 3 is 17, write 7 carry 1; thousands $7 \times 1 = 7$, plus 1 is 8. $\boxed{8736 \text{ seats}}$

Solution (b): 8736 lies between 8000 and 9000; its hundreds digit is 7, which is 5 or more, so it rounds up. $\boxed{9000}$

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9. Multiply 5,704 × 9.

Solution: Ones: $9 \times 4 = 36$; write 6, carry 3. Tens: $9 \times 0 = 0$, plus 3 is 3. Hundreds: $9 \times 7 = 63$; write 3, carry 6. Thousands: $9 \times 5 = 45$, plus 6 is 51. The zero in the tens place must keep its column — dropping it turns the problem into 574×9 and loses a whole place value. $\boxed{51336}$

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10. Partial products for 3,208 × 6.

Solution: $3{,}208 = 3000 + 200 + 0 + 8$, so one of the four parts is zero:

$$6 \times 3000 = \boxed{18000}, \quad 6 \times 200 = \boxed{1200}, \quad 6 \times 0 = \boxed{0}, \quad 6 \times 8 = \boxed{48}$$

Total: $18000 + 1200 + 0 + 48 = 19248$. Writing the zero part explicitly is worth the ink: it is the place-value column the standard algorithm would otherwise be tempted to skip. $\boxed{19248}$

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11. Bricks: 1,875 per pallet, 4 pallets. (a) Round. (b) Estimate. (c) Exact. (d) Reasonable?

Solution (a): 1875 is between 1000 and 2000; its hundreds digit 8 is 5 or more, so it rounds up. $\boxed{2000}$

Solution (b): $4 \times 2000 = 8000$. $\boxed{8000}$

Solution (c): 4×1875: ones $4 \times 5 = 20$, write 0 carry 2; tens $4 \times 7 = 28$, plus 2 is 30, write 0 carry 3; hundreds $4 \times 8 = 32$, plus 3 is 35, write 5 carry 3; thousands $4 \times 1 = 4$, plus 3 is 7. $\boxed{7500 \text{ bricks}}$

Solution (d) — instructor-judged. A full-credit answer compares the two numbers and uses the direction of the rounding. Model answer: "The estimate was 8000 and the exact answer is 7500, which is close to it, so the answer is reasonable. It should be a bit *under* the estimate, because I rounded 1875 up to 2000." Answering only "yes" or restating 7500 is not full credit.

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12. 1,608 × 5: (a) the product, (b) why the tens digit of the product is not zero.

Solution (a): Ones: $5 \times 8 = 40$; write 0, carry 4 tens. Tens: $5 \times 0 = 0$, plus the 4 carried tens is 4. Hundreds: $5 \times 6 = 30$; write 0, carry 3. Thousands: $5 \times 1 = 5$, plus 3 is 8. $\boxed{8040}$

Solution (b) — instructor-judged. A full-credit answer says where the tens digit came from. Model answer: "5×0 tens really is 0 tens, but the ones step gave 40, so 4 tens were regrouped into the tens column. $0 + 4 = 4$, so the tens digit of the product is 4." An answer that says the zero can be ignored, or that gives no source for the 4, is not full credit — the point is that a zero digit holds its column open for a carry.